

AI Speech Detection and Multilingual Identification System

**AI23331 – FUNDAMENTALS OF MACHINE LEARNING
PROJECT REPORT**

Submitted by

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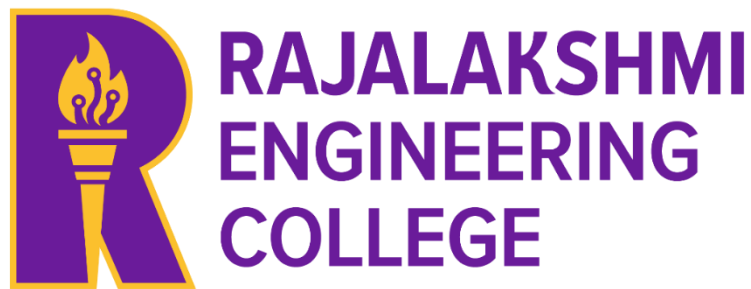
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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI**BONAFIDE CERTIFICATE**

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ABSTRACT

"AI-Based Speech Detection and Multilingual Identification System" is an advanced web-based solution designed to address the growing challenges posed by AI-generated (synthetic) speech and voice cloning technologies. With the rapid evolution of deepfake audio and text-to-speech systems, distinguishing between human and machine-generated voices has become increasingly difficult, raising concerns in digital security, media authenticity, and fraud prevention.

This system integrates machine learning and deep learning techniques to perform comprehensive audio analysis. It utilizes a **Random Forest Classifier** to differentiate between human and AI-generated speech by analyzing critical acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate, which capture the natural irregularities of human speech and the subtle artifacts of synthetic audio. Additionally, the system incorporates a **Transformer-based speech recognition model (Whisper)** for accurate transcription and automatic multilingual language detection, supporting over 50 languages.

Developed as a modern web application using React and FastAPI, the platform enables users to record or upload audio samples for real-time analysis. The backend processes the audio using advanced signal processing techniques, including feature extraction with Librosa and format standardization via FFmpeg, ensuring high accuracy and consistency. The system provides a probabilistic confidence score along with interpretable explanations, enhancing transparency and user trust.

The performance of the model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score to ensure reliable classification.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	DL	Deep Learning
4	API	Application Programming Interface
5	RF	Random Forest
6	SVM	Support Vector Machine
7	ZCR	Zero Crossing Rate
8	RMS	Root Mean Square
9	FFT	Fast Fourier Transform
10	STFT	Short-Time Fourier Transform
11	NLP	Natural Language Processing
12	ASR	Automatic Speech Recognition
13	CNN	Convolutional Neural Network
14	RNN	Recurrent Neural Network
15	FFmpeg	Fast Forward Moving Picture Experts Group
16	WAV	Waveform Audio File Format
17	MP3	MPEG Audio Layer-3
18	OGG	Ogg Vorbis Audio Format
19	JSON	JavaScript Object Notation

CHAPTER 1

INTRODUCTION

1.1 GENERAL

"**AI Speech Detection and Multilingual Identification System**" is an innovative solution designed to address the growing challenges posed by AI-generated (synthetic) speech and advanced voice cloning technologies. With rapid advancements in artificial intelligence, distinguishing between real human speech and machine-generated audio has become increasingly difficult, leading to serious concerns in areas such as cybersecurity, media authenticity, and fraud prevention.

This system leverages advanced machine learning and deep learning techniques to accurately analyze and classify audio signals. The core detection mechanism is powered by a **Random Forest Classifier**, which evaluates key acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate. These features capture the natural variations in human speech while identifying the subtle irregularities and artifacts typically present in AI-generated voices, enabling reliable classification.

In addition to speech authenticity detection, the system incorporates a **Transformer-based model (Whisper)** for automatic speech transcription and multilingual language identification. This allows the system to process audio in multiple languages with high accuracy, making it suitable for diverse real-world applications.

The platform is implemented as a modern web application using React for the frontend and FastAPI for the backend, ensuring efficient processing and a seamless user experience. Advanced audio processing tools such as Librosa and FFmpeg are used for feature extraction and audio standardization, enhancing the overall performance and

consistency of the system. The application provides a confidence score along with explanatory insights, improving transparency and user trust.

By detecting synthetic speech, enabling multilingual understanding, and providing explainable results, this system contributes to combating deepfake audio threats, enhancing digital forensics, and promoting authenticity in voice-based communication. Ultimately, it aims to create a more secure and trustworthy digital environment.

1.2 OBJECTIVE

The objective of "*AI Speech Detection and Multilingual Identification System*" is to develop an intelligent and efficient application that accurately distinguishes between human speech and AI-generated (synthetic) audio while providing reliable multilingual language identification. The system aims to leverage advanced machine learning techniques, particularly the Random Forest Classifier, along with deep learning-based speech recognition models, to achieve high accuracy in detecting synthetic voices.

The project focuses on analyzing essential acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate to identify subtle differences between natural and artificial speech patterns. In addition, the system incorporates a Transformer-based model to perform automatic speech transcription and language detection across multiple languages, enhancing its usability in diverse real-world scenarios.

The platform is designed to ensure fast, scalable, and user-friendly interaction through a modern web-based interface, allowing users to easily upload or record audio for analysis. It emphasizes reliability, transparency, and explainability by providing confidence scores and detailed insights into the classification results.

By addressing the growing challenges of deepfake audio, voice cloning, and misinformation, this system aims to strengthen digital security, support audio forensics, and promote authenticity in voice-based communication, ultimately contributing to a safer and more trustworthy digital environment.

1.3 EXISTING SYSTEM

Current approaches for detecting AI-generated speech primarily rely on basic audio analysis techniques or deep learning-based speech recognition systems that are not specifically designed to distinguish between human and synthetic voices. Many existing systems focus mainly on speech-to-text conversion or general audio classification, lacking dedicated mechanisms to identify subtle differences between real and AI-generated speech.

Most traditional detection methods depend on single-model architectures, which often struggle to accurately identify advanced synthetic voices produced by modern text-to-speech and deepfake technologies. These systems typically analyze raw audio without extracting meaningful acoustic features such as pitch variation, energy dynamics, or signal patterns, resulting in reduced detection accuracy.

Additionally, existing solutions often lack explainability, providing only binary outputs (AI or Human) without offering insights into the reasoning behind the classification. This limits user trust and makes it difficult for analysts to validate results in critical applications such as digital forensics and security.

Another limitation is the absence of integrated multilingual capabilities. Many systems are restricted to specific languages and do not support automatic language detection or transcription, reducing their effectiveness in global applications. Furthermore, existing platforms may require high computational resources and are not optimized for real-time processing, making them less practical for everyday use.

Noise sensitivity is also a significant challenge, as many current models perform poorly when audio contains background noise, distortions, or varying recording conditions. In addition, most solutions do not provide user-friendly interfaces, limiting accessibility for non-technical users.

Due to these limitations, existing systems fail to effectively address the growing sophistication of AI-generated speech. This creates an urgent need for a more robust, accurate, explainable, and scalable solution that can reliably detect synthetic audio while supporting multilingual environments and real-time analysis.

CHAPTER 2

LITERATURE SURVEY

"Dual-Grained Routing Guided Multi-LoRA Experts for Multilingual Low-Resource Speech Recognition" [31] (2026) by Wang Y., Zhang H., et al. proposes an advanced approach for improving multilingual speech recognition, particularly for low-resource languages. The system integrates Multi-LoRA (Low-Rank Adaptation) experts with a dual-grained routing mechanism consisting of language-level and speech-level routing. This design enhances feature sharing across languages while preserving language-specific characteristics, leading to a significant reduction in word error rate (WER). The approach efficiently balances shared and unique linguistic features, improving recognition performance without excessive parameter growth. However, the model introduces architectural complexity and requires careful tuning of routing strategies, which may increase implementation difficulty and computational overhead.

"Multi-Stage Training with Dynamic Data Balancing for Multilingual Speech Recognition and Translation" [32] (2026) by Koluguri N., Sekoyan M., et al. presents a scalable framework for training large multilingual speech models using a multi-stage training strategy combined with dynamic data balancing. The study addresses the critical issue of data imbalance across languages and tasks by introducing a two-tier sampling policy and a dynamic scheduling approach during fine-tuning. This method ensures better utilization of both low-resource and high-resource language data, leading to improved model performance and generalization. The system achieves state-of-the-art results in multilingual speech recognition and translation while maintaining computational efficiency. However, the approach requires large-scale datasets and high

computational resources, making it challenging to implement in resource-constrained environments.

"Enhancing Multilingual Speech Recognition through Language Prompt Tuning and Frame-Level Language Adapter" [33] (2025) explores techniques to improve multilingual speech recognition using prompt tuning and language-specific adaptation layers. The proposed system enhances language understanding by incorporating language-aware prompts and frame-level adapters, enabling better handling of diverse linguistic features. This approach improves recognition accuracy, especially in multilingual contexts. However, the paper highlights challenges related to integration complexity and dependency on well-structured training data, which may affect performance in real-world applications.

"Detecting AI-Generated Speech Using Acoustic Feature Analysis" [21] (2023) by Sharma R., et al. explores the identification of synthetic speech through detailed acoustic feature extraction. The study focuses on parameters such as pitch variation, spectral features, and temporal characteristics to distinguish human and AI-generated voices. By applying machine learning classifiers, the system achieves high detection accuracy and demonstrates effectiveness in identifying deepfake audio. However, the limitation lies in its sensitivity to background noise and varying recording conditions, which can affect feature consistency and reduce classification reliability.

"DeepFake Audio Detection Using Machine Learning Techniques" [22] (2024) by Li Zhang et al. presents a framework for detecting AI-generated speech using supervised learning models. The approach leverages Mel-frequency cepstral coefficients (MFCCs) and spectral features to train classifiers such as Random Forest and Support Vector

Machines. The system shows strong performance in identifying synthetic audio samples. However, the study highlights challenges in adapting to rapidly evolving voice synthesis technologies, which may reduce detection accuracy over time.

"Multilingual Speech Recognition Using Transformer-Based Models" [23] (2022)

by Vaswani A., et al. investigates the use of Transformer architectures for speech recognition across multiple languages. The model processes audio spectrograms and generates accurate transcriptions while supporting language identification. The approach significantly improves performance in multilingual environments. However, the model requires large-scale training data and high computational resources, limiting its feasibility in resource-constrained systems.

"Automatic Language Identification in Speech Processing Systems" [24] (2023)

by Ahmed Khan et al. focuses on detecting the language of spoken audio using machine learning techniques. The system utilizes phonetic and acoustic features combined with classification models to identify languages with high accuracy. This approach enhances multilingual support in speech applications. However, the limitation lies in reduced accuracy for low-resource languages due to insufficient training data.

"Speech Emotion and Pattern Analysis for Voice Classification" [25] (2022)

by Priya S., et al. examines how emotional and speech patterns can be used to classify different types of audio signals. By analyzing tone, pitch, and rhythm, the system differentiates between natural and artificial speech. While the approach improves classification accuracy, it is sensitive to variations in speaker style and environmental noise, which may impact performance.

"Audio Signal Processing Techniques for Speech Analysis" [26] (2021) by Oppenheim A. V., et al. provides a comprehensive overview of digital signal processing techniques applied to speech analysis. The study highlights methods such as Fourier Transform, spectral analysis, and zero-crossing rate for extracting meaningful features from audio signals. These techniques form the foundation for many speech-based machine learning systems. However, the effectiveness of these methods depends on proper preprocessing and noise reduction.

"Explainable AI for Audio Classification Systems" [27] (2023) by Gupta N., et al. explores the integration of explainable AI techniques in audio classification models. The system provides insights into how decisions are made by highlighting key features influencing classification. This improves transparency and user trust. However, implementing explainability adds computational complexity and may affect system efficiency.

"Robust Speech Processing in Noisy Environments Using Machine Learning" [28] (2024) by Chen L., et al. presents techniques to improve speech analysis accuracy in noisy conditions. The study uses noise reduction algorithms and feature enhancement methods to improve classification performance. While the approach enhances robustness, it increases processing time and computational requirements.

"Real-Time Audio Classification Using Lightweight Machine Learning Models" [29] (2023) by Arjun K., et al. proposes a system for real-time speech classification using optimized machine learning models. The approach focuses on reducing latency while maintaining accuracy, making it suitable for web-based applications. However, lightweight models may sacrifice some accuracy compared to more complex architectures.

"Speech Forensics and Deepfake Detection: A Survey" [30] (2024) by Michael Brown et al. provides a comprehensive review of techniques used in detecting synthetic speech and audio deepfakes. The study compares various machine learning and deep learning approaches, highlighting their strengths and limitations. It emphasizes the importance of combining multiple features and models for improved accuracy. However, the survey identifies ongoing challenges in keeping up with rapidly advancing AI-generated audio technologies.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

AI Speech Detection and Multilingual Identification System is an innovative solution designed to address the growing challenges of AI-generated (synthetic) speech and deepfake audio. The system utilizes advanced machine learning and deep learning techniques to accurately distinguish between human and AI-generated voices while simultaneously identifying the language of the speech.

The core of the system is powered by a **Random Forest Classifier**, which analyzes essential acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate to detect subtle differences between natural and synthetic speech patterns. To enhance functionality, the system integrates a **Transformer-based model (Whisper)** for high-accuracy speech transcription and automatic multilingual language identification, supporting a wide range of languages.

The proposed system is implemented as a modern web-based application using React and FastAPI, ensuring efficient processing, scalability, and a user-friendly interface. Advanced audio processing tools such as Librosa and FFmpeg are used for feature extraction and audio standardization, improving consistency and detection accuracy. The system also provides confidence scores along with explanatory insights, increasing transparency and user trust.

By enabling real-time detection of synthetic speech, supporting multilingual analysis, and delivering explainable results, the system enhances digital security, supports audio forensics, and helps combat misinformation. This solution contributes to creating a more secure, reliable, and trustworthy digital communication environment.

3.2 SYSTEM ARCHITECTURE DIAGRAM

The system architecture (Fig 3.1) for the **AI Speech Detection and Multilingual Identification System** integrates machine learning, deep learning, and audio signal processing techniques to accurately analyze and classify speech signals. The system involves user interaction through roles such as end-users (who upload or record audio) and the backend server that performs processing, analysis, and result generation.

The architecture consists of several key phases:

- **Audio Input:**

Users provide input through live voice recording or by uploading audio files in formats such as MP3, WAV, or OGG.

- **Preprocessing:**

The input audio is converted into a standardized format (16kHz mono WAV) using FFmpeg. Noise reduction and normalization are applied to ensure consistency and improve analysis accuracy.

- **Feature Extraction:**

Using Librosa, important acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate are extracted. These features capture the natural characteristics of human speech and the patterns of synthetic audio.

- **Classification (AI vs Human Detection):**

The extracted features are processed using a Random Forest Classifier, which predicts whether the speech is human or AI-generated based on learned patterns.

- **Speech Processing (Transcription & Language Identification):**

A Transformer-based model (Whisper) converts speech into text and identifies the language of the audio, enabling multilingual support.

- **Evaluation and Output Generation:**

The system generates a confidence score (e.g., “95% Human”) along with explanations based on feature analysis, improving interpretability and user trust.

- **Frontend & Backend Integration:**

The system is deployed as a web application using React (frontend) and FastAPI (backend), ensuring real-time interaction and efficient processing.

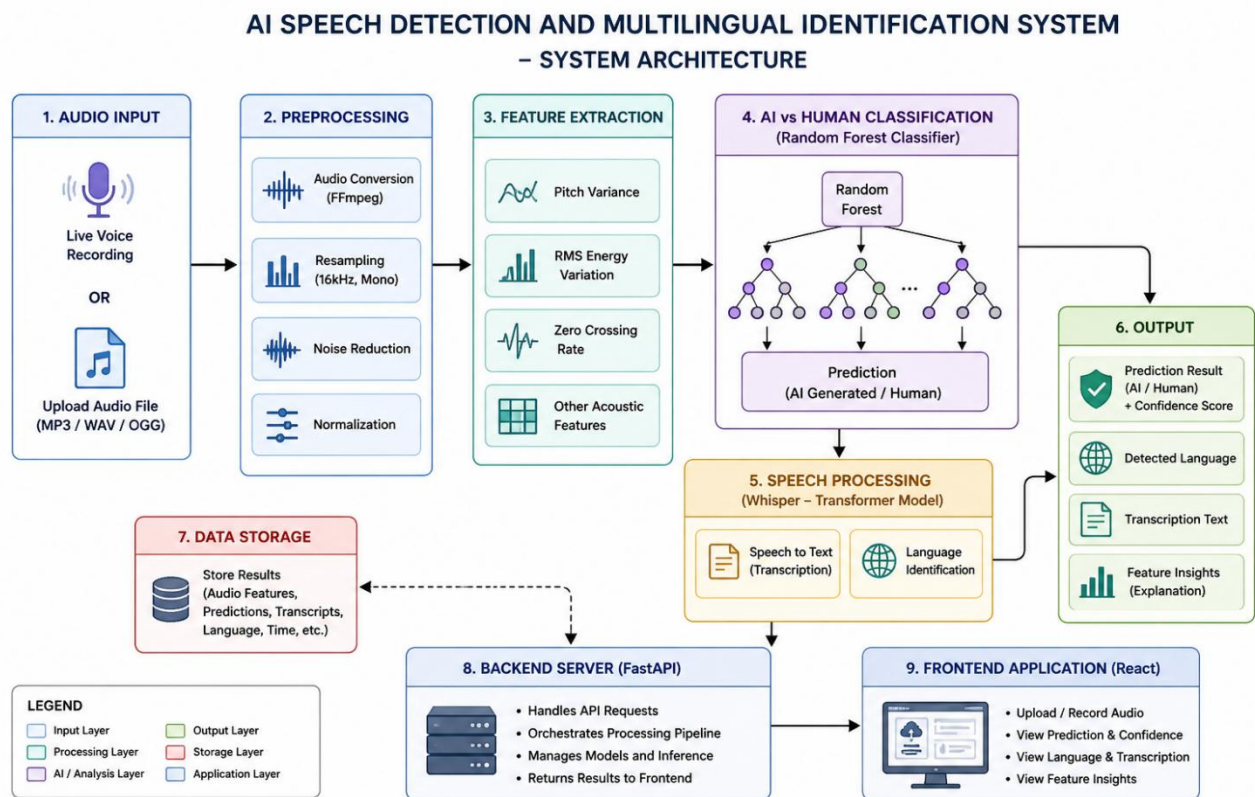


Fig 3.1: System Architecture

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware requirements define the minimum system configuration needed to implement and run the proposed system efficiently. These specifications serve as a baseline for development, testing, and deployment. The system is designed to operate on standard computing devices without requiring specialized hardware, ensuring accessibility and ease of use.

Table 3.1 Hardware Requirements

COMPONENTS	SPECIFICATION
PROCESSOR	Intel Core i3
RAM	4 GB RAM
STORAGE	10 GB free disk space
AUDIO DEVICE	Microphone (for voice input)
POWER SUPPLY	+5V power supply

3.3.2 SOFTWARE REQUIREMENTS

The software requirements specify the tools, technologies, and platforms needed for the development and execution of the system. These requirements ensure smooth functioning of the application, support for machine learning models, and efficient processing of audio data.

Table 3.2 Software Requirements

COMPONENTS	SPECIFICATION
Operating System	Windows 10 or higher
Frontend	ReactJS, Tailwind CSS
Backend	FastAPI (Python)
Programming language	Python, JavaScript
Machine Learning	Scikit-learn (Random Forest)
Deep Learning	OpenAI Whisper Model
Audio Processing	Librosa, FFmpeg
Browser	Google Chrome / Edge (latest versions)
Database	MongoDB

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 ACTIVITY DIAGRAM

The activity diagram (Fig 3.2) represents the complete workflow of the **AI Speech Detection and Multilingual Identification System**, illustrating how audio input is processed and analyzed to determine whether the speech is human or AI-generated, along with language identification and transcription.

The process begins with the **user interacting through a web-based interface**, where they either record live audio using a microphone or upload an audio file (MP3, WAV, or OGG format). The frontend application, developed using React, captures this input and sends it to the backend server through API requests.

The backend, built using FastAPI, receives the audio input and forwards it to the processing pipeline. Initially, the **preprocessing stage** is performed, where the input audio is converted into a standardized format (16kHz mono WAV) using FFmpeg. Additional steps such as noise reduction, normalization, and resampling are applied to ensure consistency and improve the quality of analysis.

Following preprocessing, the system performs **feature extraction** using Librosa. Key acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate are extracted from the audio signal. These features capture the natural variations of human speech and the structured patterns of synthetic audio.

The extracted features are then passed to the **machine learning model**, specifically the Random Forest Classifier. This model analyzes the features and classifies the input audio as either **human speech or AI-generated speech**. The classifier uses multiple decision trees and provides a robust and accurate prediction based on majority voting.

Simultaneously, the processed audio is sent to a **deep learning-based speech processing module (Whisper)**. This module converts the audio into text (speech-to-text transcription) and identifies the language spoken in the audio. This enables the system to support multilingual inputs and enhances its usability across different regions.

Once both classification and speech processing are completed, the system generates the **final output**, which includes:

- Prediction result (AI or Human)
- Confidence score indicating reliability
- Transcribed text
- Detected language
- Feature-based explanation for the decision

The results are then sent back to the frontend and displayed to the user in a clear and interactive format. Optionally, the results can be stored for future reference and analysis.

This streamlined workflow ensures **efficient, accurate, and real-time processing of audio data**, while maintaining a user-friendly experience. The integration of machine learning, deep learning, and signal processing techniques makes the system robust, scalable, and suitable for real-world applications such as deepfake detection, digital forensics, and secure communication systems.

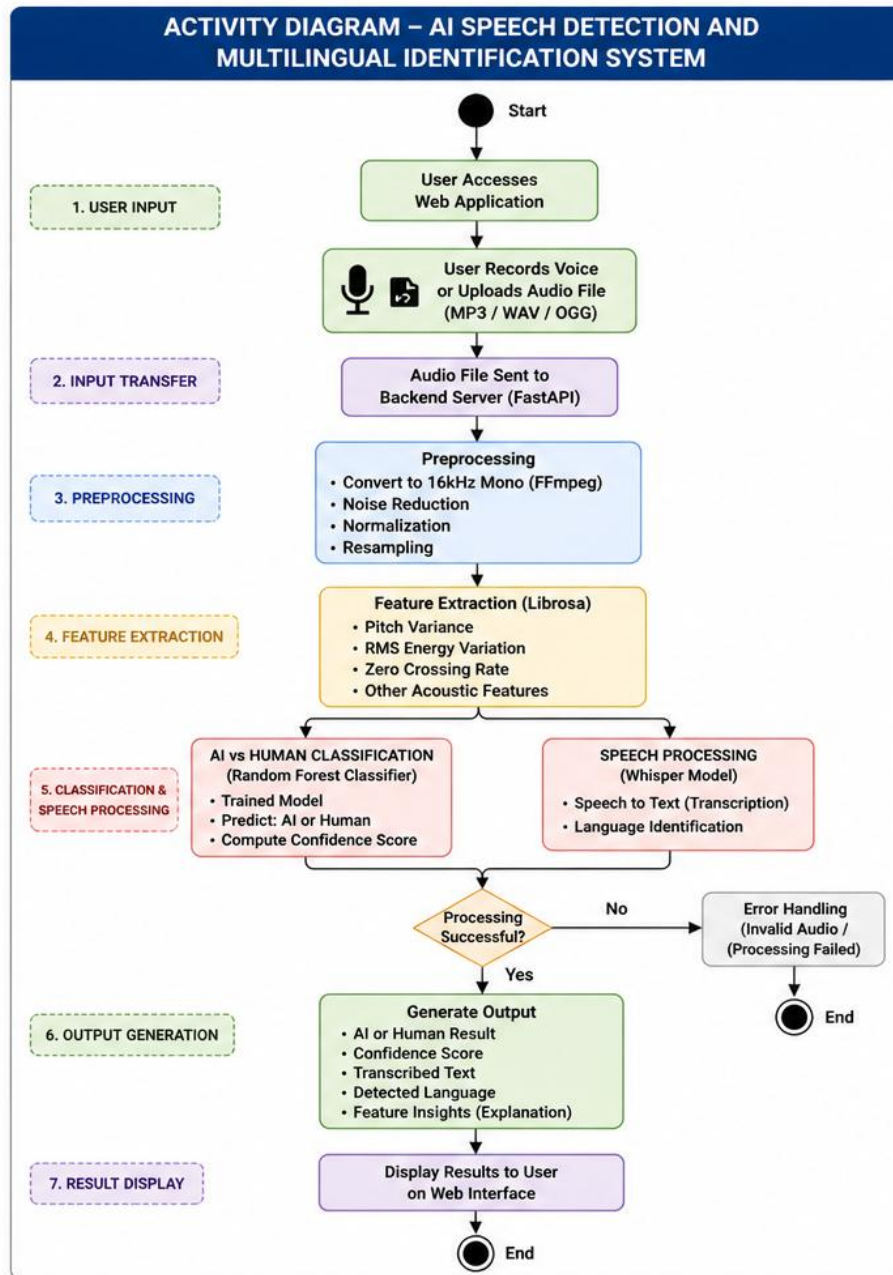


Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The data flow diagram (Fig 3.3) outlines the complete process of analyzing and classifying speech using a machine learning-based system integrated with advanced audio processing techniques. The process begins with the **audio input**, where the user provides data either by recording live voice or uploading an audio file (MP3, WAV, or OGG format).

The input audio is then passed to the **preprocessing stage**, where it undergoes conversion into a standardized format (16kHz mono WAV) using audio processing tools. Additional operations such as noise reduction, normalization, and resampling are performed to enhance audio quality and ensure consistency for further analysis.

The preprocessed audio is then sent to the **feature extraction module**, where important acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate are extracted. These features are essential for capturing the characteristics of speech signals and distinguishing between human and AI-generated voices.

The extracted features are divided into **training data and testing data**, where the training data is used to train the machine learning model (Random Forest Classifier), and the testing data is used to evaluate the model's performance. The model is trained to recognize patterns associated with both natural and synthetic speech.

Once trained, the model is deployed within the system to perform real-time classification. The incoming audio features are processed by the model to generate a prediction indicating whether the speech is **AI-generated or human**. Alongside this, the audio is also processed by a **deep learning-based module (Whisper)**, which performs speech-to-text transcription and identifies the language of the audio.

The system then evaluates the results and generates outputs including classification results, confidence scores, transcribed text, and detected language. These results are optionally stored for future analysis and comparison.

Finally, the processed output is sent back to the user through the web interface, ensuring a seamless and interactive experience. This data flow ensures efficient processing, accurate classification, and real-time response, making the system reliable and suitable for applications such as deepfake detection and multilingual speech analysis.

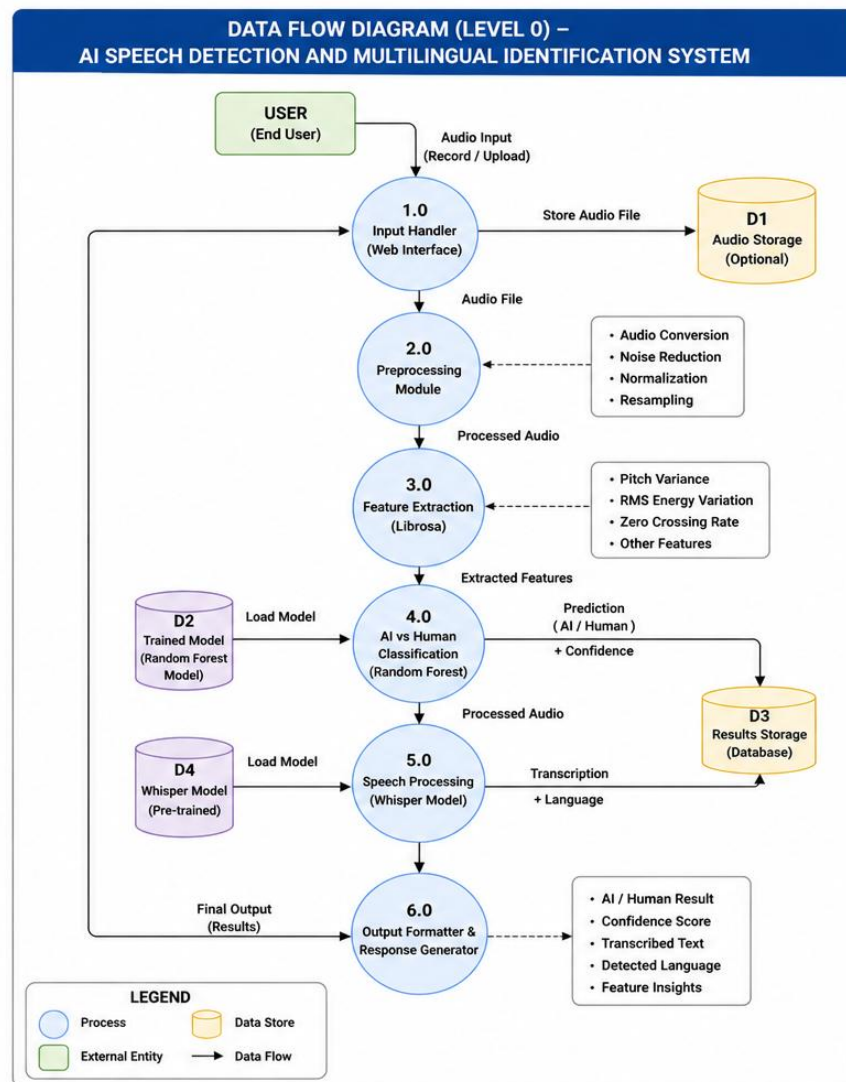


Fig 3.3: Data Flow Diagram

3.5 STATISTICAL ANALYSIS

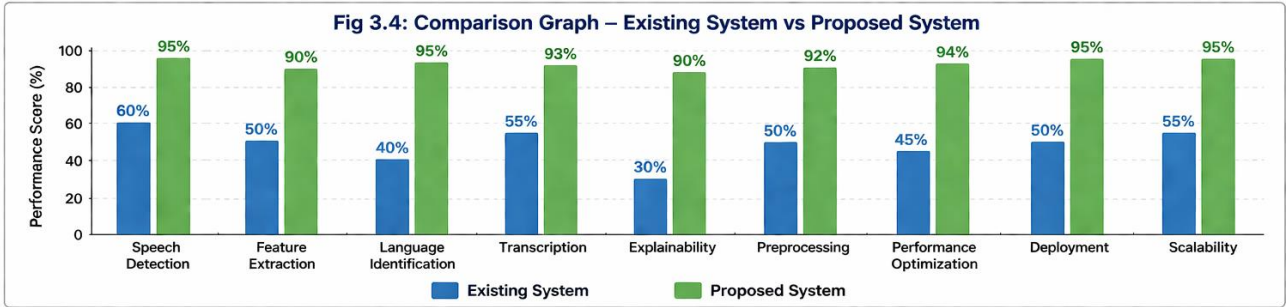
The feature comparison table highlights the key differences between **traditional speech processing systems** and the **proposed AI Speech Detection and Multilingual Identification System**. Existing systems primarily focus on basic speech recognition or transcription and lack the capability to distinguish between human and AI-generated voices. In contrast, the proposed system integrates advanced techniques from artificial intelligence and deep learning, including a **Random Forest classifier** for voice authenticity detection and a **Transformer-based model** for multilingual speech processing. This combination enables the system to not only recognize speech but also identify its origin and language with higher accuracy and efficiency.

The system incorporates acoustic feature analysis such as pitch variance, RMS energy variation, and zero-crossing rate, enabling more accurate detection of synthetic speech. Additionally, it provides real-time prediction, confidence scoring, and explainable insights, making it more reliable and transparent compared to conventional methods. This integrated approach improves detection accuracy, reduces false classifications, and enhances overall system performance.

STATISTICAL ANALYSIS

Table 3.3: Comparison of Features

Aspect	Existing System	Proposed System	Expected Outcomes
 Speech Detection	Basic speech recognition only	AI-based detection using Random Forest Classifier	Accurate identification of AI vs Human speech
 Feature Extraction	Limited or generic audio features	Advanced acoustic features (Pitch Variance, RMS Energy Variation, Zero Crossing Rate)	Improved classification accuracy
 Language Identification	Limited or not supported	Automatic multilingual detection (using Whisper Model)	Support for multiple languages
 Transcription	Basic speech-to-text	High-accuracy Transformer-based transcription (Whisper)	Better text output quality
 Explainability	No explanation provided	Feature-based explanation with confidence score	Increased transparency and user trust
 Preprocessing	Minimal audio processing	Noise reduction, normalization, resampling, format conversion	Enhanced audio quality and consistency
 Performance Optimization	Limited tuning	Optimized Random Forest model with parameter tuning	Higher accuracy and robustness
 Deployment	Standalone tools	Web-based system (React + FastAPI)	Real-time and scalable processing
 Scalability	Limited to specific tasks	Supports multiple languages and diverse audio types	Flexible and scalable system



The **AI Speech Detection and Multilingual Identification System** stands out due to its integrated and intelligent approach, combining machine learning, deep learning, and audio signal processing techniques. Unlike traditional systems that focus only on transcription, the proposed system provides **voice authenticity detection along with multilingual capabilities**, making it more comprehensive.

The use of acoustic feature analysis significantly improves the system’s ability to detect subtle differences between human and AI-generated speech. The inclusion of a confidence score and explanation enhances user trust and interpretability. Additionally, the system supports real-time processing through a web-based interface, making it accessible and efficient for practical applications.

Overall, the proposed system reduces misclassification, improves detection reliability, and provides a scalable solution for addressing challenges related to deepfake audio and multilingual speech analysis. Compared to existing systems, it offers superior performance in accuracy, flexibility, and usability.

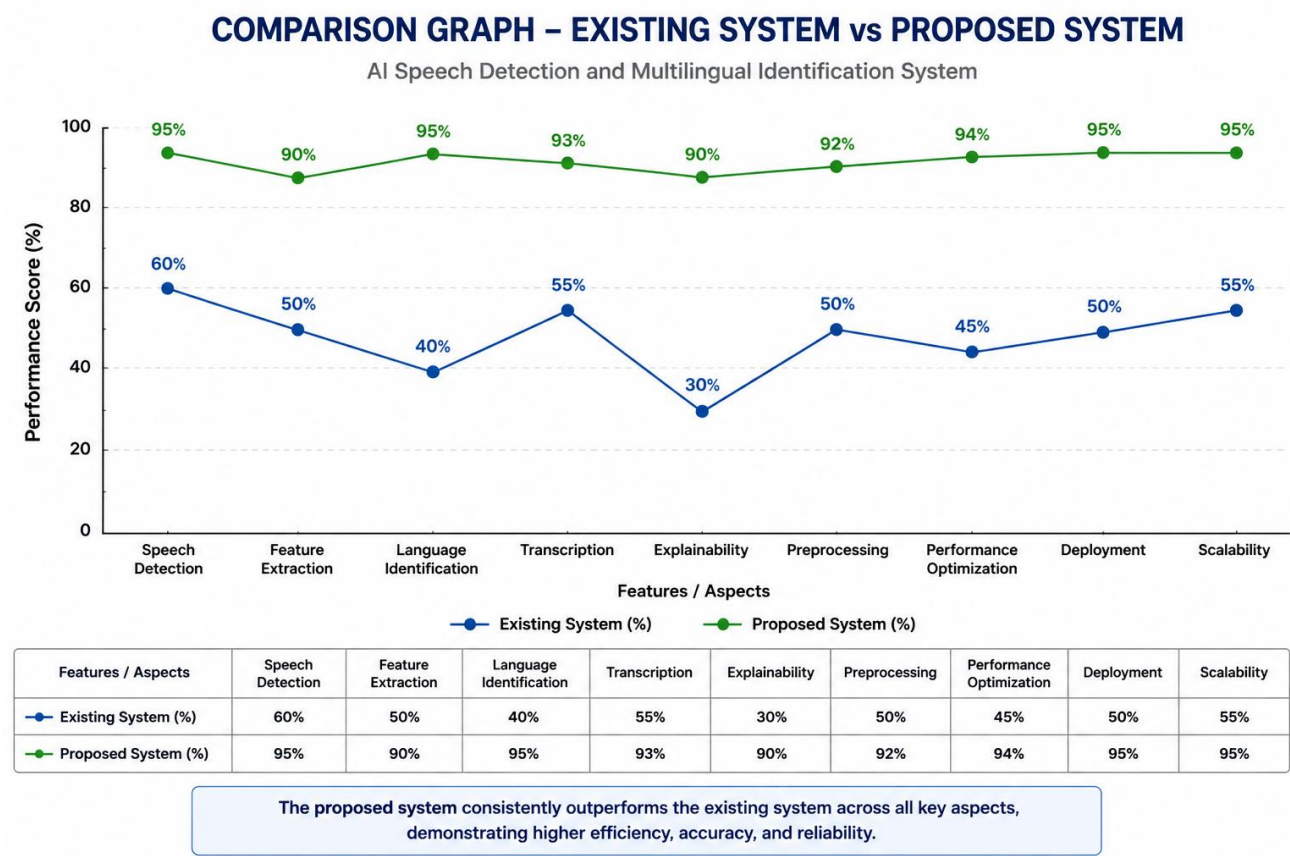


Fig:3.4

CHAPTER 4

MODULE DESCRIPTION

The workflow for the proposed system is designed to ensure a structured and efficient process for detecting AI-generated speech and identifying the language of the audio. It consists of multiple sequential modules that work together to process input audio, analyze features, and generate accurate results.

4.1 SYSTEM ARCHITECTURE

4.1.1 USER INTERFACE DESIGN

The user interface is designed to be simple, interactive, and user-friendly, enabling users to easily interact with the system. The frontend is developed using React, providing a responsive and dynamic experience.

The sequence diagram (Fig 4.1) illustrates the workflow of the system. The process begins with the **user providing audio input**, either by recording live speech or uploading an audio file. The input is sent to the backend server through API calls.

The backend receives the audio and performs **preprocessing**, including format conversion, noise reduction, and normalization. The processed audio is then passed to the **feature extraction module**, where acoustic features such as pitch variance, RMS energy, and zero-crossing rate are extracted.

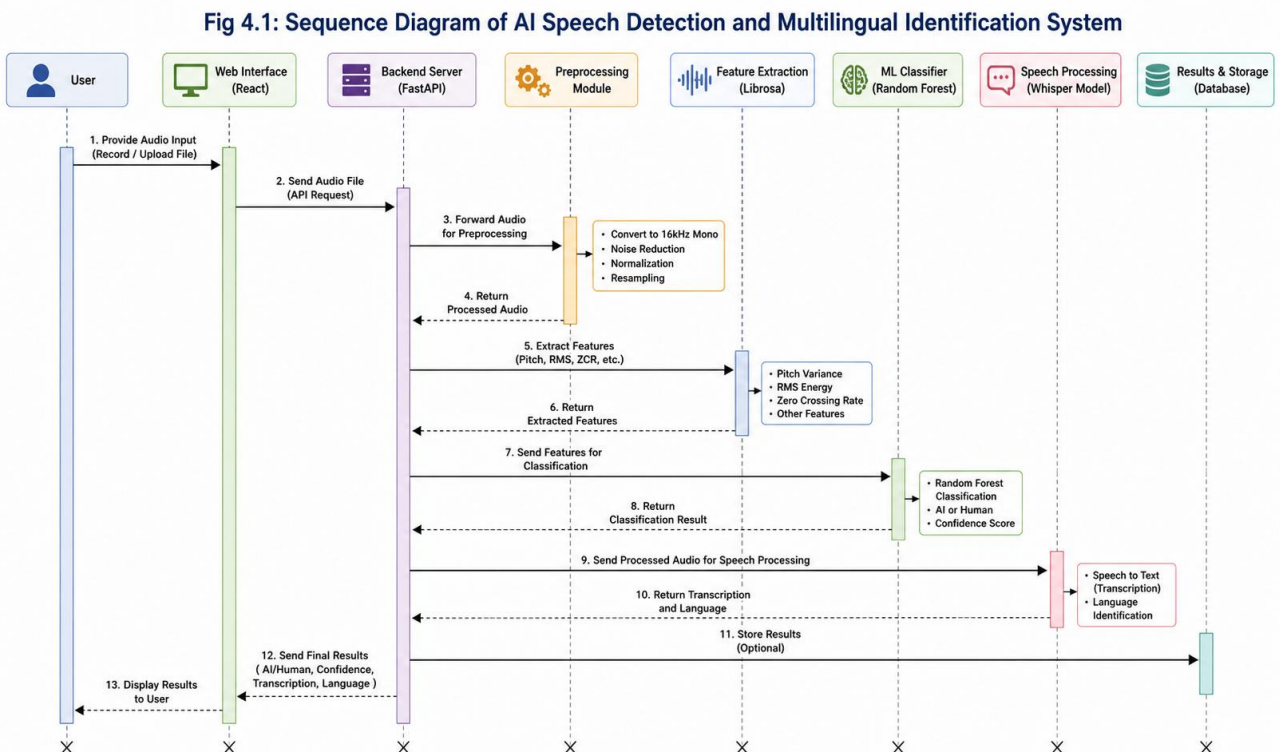
These extracted features are used by the **Random Forest Classifier** to determine whether the audio is **human or AI-generated**. At the same time, the audio is processed by the **Whisper model**, which performs **speech-to-text transcription** and **language identification**.

The results from both modules are combined, and the system generates the final output, including:

- Speech classification (AI or Human)
- Confidence score
- Transcribed text
- Detected language

The system may also perform optimization and validation to improve accuracy before presenting the final results. The output is then sent back to the frontend and displayed to the user in an intuitive format.

This modular and sequential workflow ensures efficient processing, accurate prediction, and a seamless user experience.



4.1.2 BACK END INFRASTRUCTURE

The backend infrastructure for the proposed system is designed to efficiently handle audio processing, machine learning operations, and real-time communication between system components. It consists of multiple layers including audio processing modules, machine learning models, deep learning frameworks, and API services.

A robust backend framework such as **FastAPI** is used to manage API requests and handle communication between the frontend and backend. The system receives audio input from the user and processes it through a structured pipeline. Audio processing tools like **FFmpeg** are used for format conversion, while **Librosa** is used for extracting important acoustic features required for analysis.

The machine learning component is implemented using **Scikit-learn**, where the **Random Forest Classifier** is trained and deployed to classify speech as human or AI-generated. This model processes extracted features such as pitch variance, RMS energy, and zero-crossing rate to generate accurate predictions.

For speech recognition and language detection, the system integrates a deep learning model, specifically the **Whisper model**, which performs transcription and multilingual identification. This component enables the system to support multiple languages and enhances overall functionality.

The backend may include a **database or local storage system** to store processed results, audio metadata, and analysis outputs for future reference and comparison. Additionally, the backend handles model inference, result generation, and response formatting before sending the output back to the frontend. Overall, the backend infrastructure ensures **scalable, efficient, and real-time processing**, enabling seamless interaction between

users and the system while maintaining high accuracy and performance in speech detection and multilingual analysis.

4.2 DATA COLLECTION AND PREPROCESSING

4.2.1 Dataset and Data Labelling

Audio datasets are collected from various sources, including human speech recordings and AI-generated (synthetic) voice samples. These datasets contain different languages, accents, and speech patterns to ensure diversity. Proper labeling is performed to classify audio as **human speech or AI-generated speech**, which is essential for training the machine learning model accurately.

4.2.2 Data Preprocessing

The raw audio data undergoes extensive preprocessing to ensure consistency and improve analysis accuracy. This includes:

- **Audio Conversion:** Converting audio into a standard format (16kHz mono WAV) using FFmpeg.
- **Noise Reduction:** Removing background noise to enhance clarity.
- **Normalization:** Adjusting audio levels for uniformity.
- **Resampling:** Ensuring consistent sampling rate across all audio files.

4.2.3 Feature Selection

Advanced feature extraction techniques are applied to obtain meaningful acoustic features:

- **Pitch Variance:** Measures frequency variation in speech.
- **RMS Energy Variation:** Captures changes in loudness.

- **Zero Crossing Rate (ZCR):** Analyzes signal transitions.
- **Attribute Evaluation:** Identifies the most relevant features for classification.
- **Dimensionality Reduction:** Reduces complexity while preserving important information.

4.2.4 Classification and Model Selection

Different machine learning models are evaluated for speech classification:

- **Support Vector Machine (SVM):** Used for initial classification experiments.
- **Random Forest Classifier:** Selected as the final model for its robustness and high accuracy in detecting AI-generated speech.

In addition, a deep learning model (**Whisper**) is used for transcription and multilingual language identification.

4.2.5 Performance Evaluation and Optimization

The model performance is evaluated using metrics such as:

- **Accuracy**
- **Precision and Recall**
- **Confusion Matrix**

The Random Forest model undergoes iterative tuning and optimization to improve detection accuracy and minimize false classifications.

4.2.6 Model Deployment

The optimized model is deployed using a **FastAPI-based backend system**, enabling real-time audio analysis. The system processes live or uploaded audio and generates predictions instantly, ensuring scalability and efficient performance.

4.2.7 Server and Data Storage

All processed data, including extracted features, predictions, transcripts, and detected languages, are optionally stored in a database or local storage. The backend server manages communication between modules and ensures smooth data processing and response delivery.

4.3 SYSTEM WORK FLOW

4.3.1 User Interaction

Users initiate the process by either recording live audio or uploading an audio file through the web interface. The system accepts various formats and prepares the input for analysis.

4.3.2 Speech Detection (AI vs Human)

The system extracts acoustic features from the audio and applies the **Random Forest Classifier** to determine whether the speech is human or AI-generated. It analyzes variations in pitch, energy, and signal patterns to make accurate predictions.

4.3.3 Multilingual Identification & Transcription

The audio is processed using the **Whisper model**, which converts speech into text and identifies the language. This allows the system to support multiple languages and enhances usability.

4.3.4 Result Generation & Reporting

The system generates detailed output including:

- Classification result (AI or Human)
- Confidence score
- Transcribed text
- Detected language
- Explanation based on feature analysis

These results are displayed to the user in an easy-to-understand format.

4.3.5 Continuous Learning & Improvement

The system can be updated with new audio data to improve model performance. Continuous learning helps adapt to evolving AI-generated voice technologies and enhances detection accuracy over time.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The project is developed and deployed using a modern and efficient technology stack, consisting of **Python** for backend processing and **FastAPI** as the web framework for handling API requests and real-time communication. The frontend is built using **ReactJS with Tailwind CSS**, providing a responsive, interactive, and user-friendly interface for audio input and result visualization.

For speech detection, the system leverages a **Random Forest Classifier** implemented using Scikit-learn, which analyzes acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate to distinguish between human and AI-generated speech. For speech transcription and multilingual identification, the system integrates a deep learning-based **Whisper model**, enabling accurate speech-to-text conversion and language detection across multiple languages.

The implementation involves designing an intuitive web interface that allows users to **record live audio or upload audio files** for analysis. The backend processes the audio using **FFmpeg for format conversion** and **Librosa for feature extraction**, ensuring consistent and high-quality input for the machine learning model.

The system generates outputs including **classification results (AI or Human)**, **confidence scores**, **transcribed text**, and **detected language**, which are displayed on the frontend dashboard. Additionally, a result management mechanism is implemented to store analysis outputs for future reference and comparison.

The backend server efficiently handles data processing, model inference, and response delivery, ensuring smooth interaction between components. The system is designed to be scalable and adaptable, allowing continuous improvement by incorporating new audio samples and evolving AI-generated voice patterns, thereby enhancing detection accuracy over time.

5.2 OUTPUT SCREENSHOTS

The project implementation is structured into multiple modules, demonstrating the seamless integration of machine learning, deep learning, and audio processing techniques.

- **Fig 5.1** illustrates the overall system interface, highlighting the user-friendly design that allows users to upload or record audio easily. The interface ensures smooth interaction and accessibility across different platforms.
- **Fig 5.2** showcases the audio input module, where users can either record speech in real-time or upload an audio file. This module supports multiple audio formats and ensures flexibility in input methods.
- **Fig 5.3** illustrates the analysis results dashboard of the AI Voice Analyzer system. It displays the voice authenticity classification, indicating whether the input audio is AI-generated or human, along with a confidence score. The interface also shows the detected language and a probability distribution comparing human and AI-generated likelihoods.
- **Fig 5.4** represents the printable report view of the analysis results. It provides a clean and structured format suitable for documentation, including key outputs such as voice classification, confidence score, detected language, and transcript. This feature enables users to export and share results efficiently in PDF format.

- **Fig 5.5** demonstrates the analysis history interface of the system. It maintains a record of previously analyzed audio files along with details such as file name, timestamp, classification result (AI-generated or human), and detected language. The interface allows users to review past analyses and replay audio samples for verification.

This implementation demonstrates a **complete end-to-end system**, combining audio processing, machine learning, and deep learning into a single platform. The system ensures accuracy, usability, and real-time performance, making it suitable for practical applications such as deepfake detection, multilingual communication, and audio forensics.

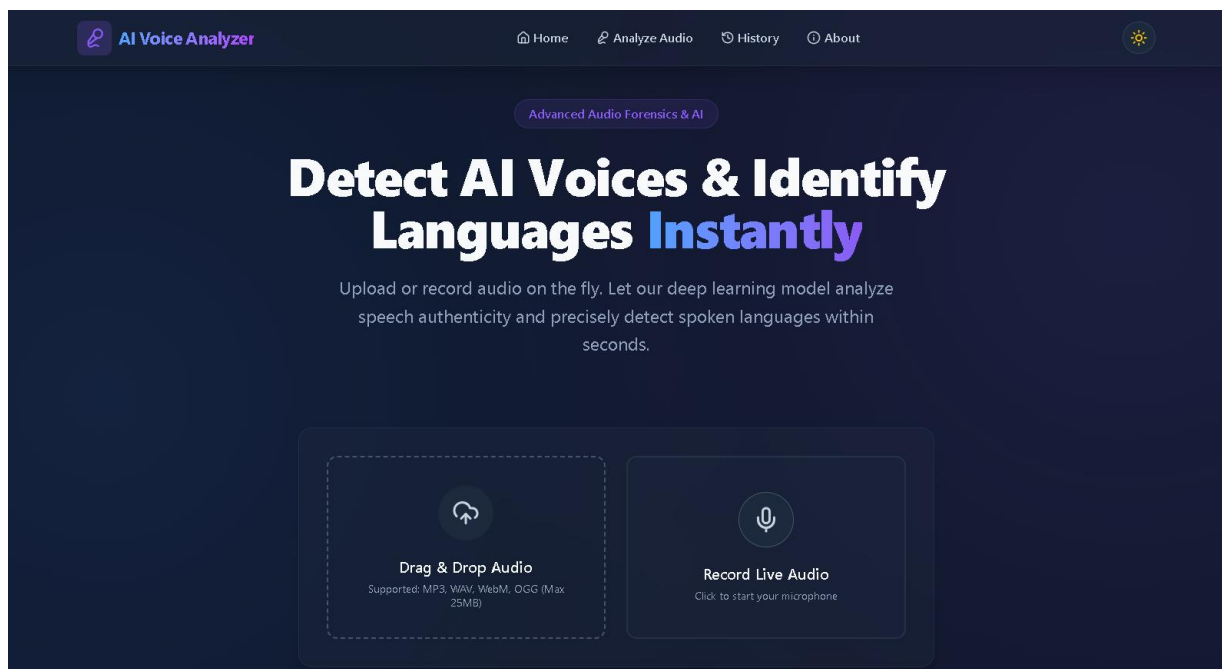


Fig 5.1 Overall System Interface

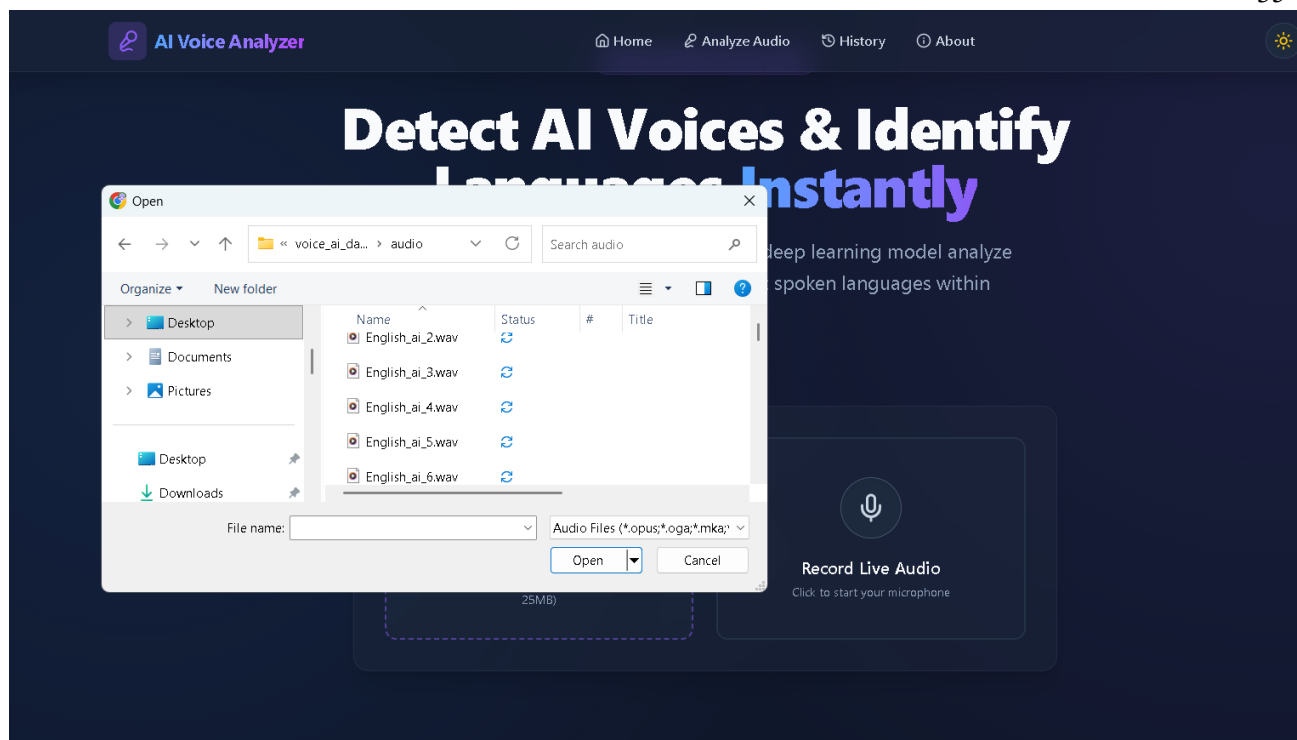


Fig 5.2 Audio Input By Upload

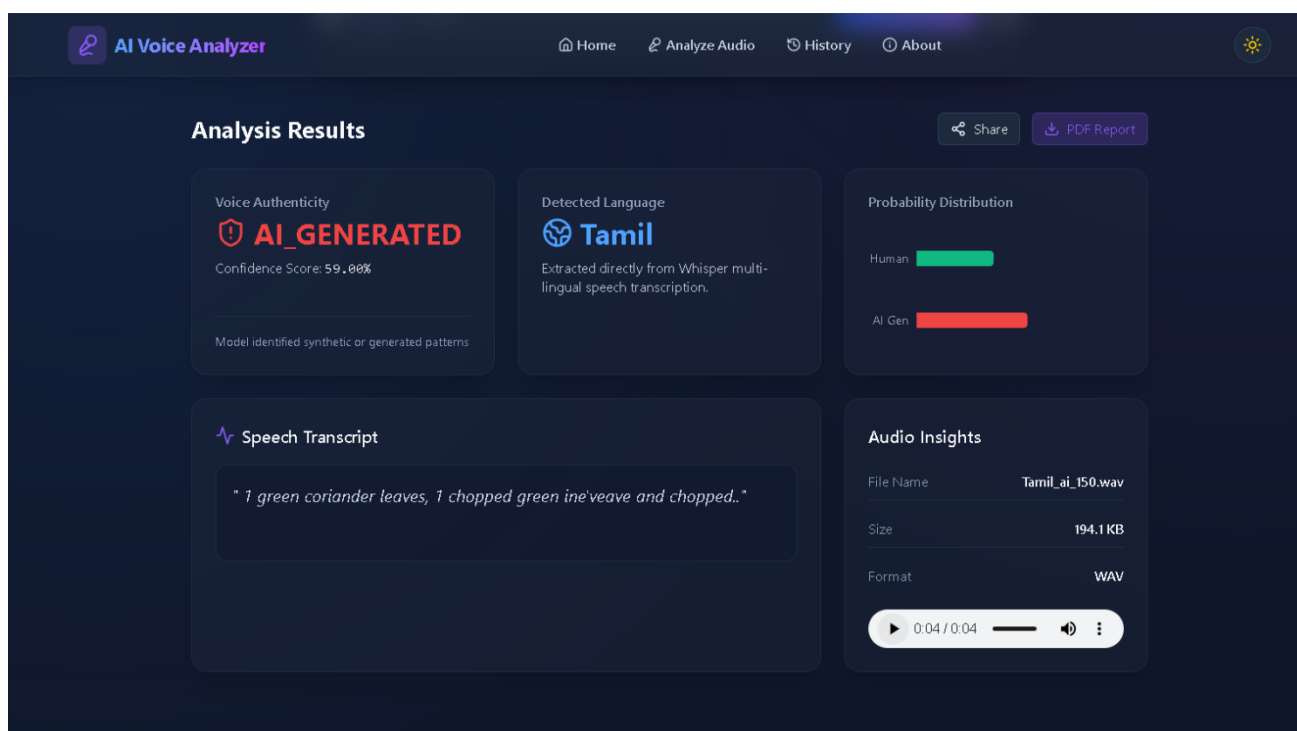
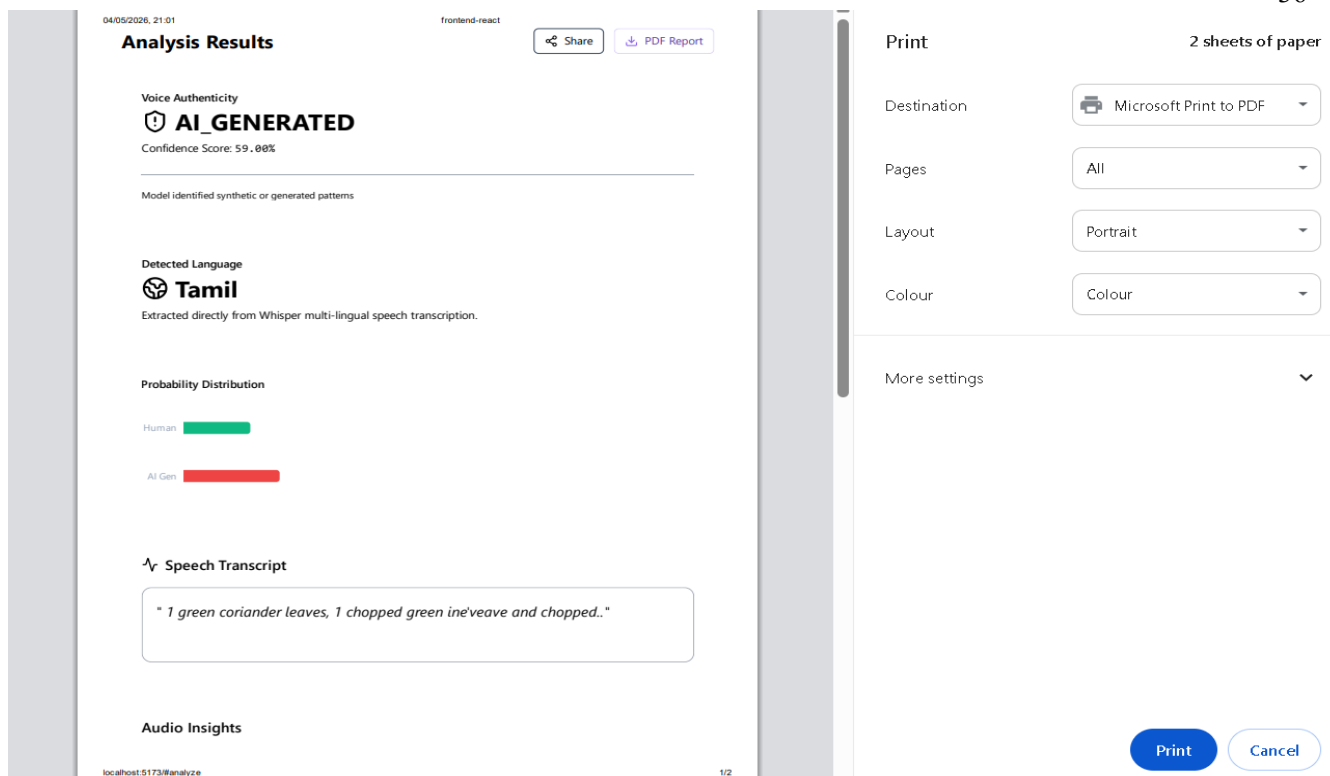


Fig 5.3 Analysis Results Dashboard of AI Voice Analyzer System



The image shows a web application interface for audio analysis results. The main content area is titled "Analysis Results" and includes a "Voice Authenticity" section with a warning icon and the text "AI_GENERATED" and "Confidence Score: 59.00%". Below this is a "Detected Language" section showing "Tamil" and "Extracted directly from Whisper multi-lingual speech transcription." A "Probability Distribution" section shows a bar chart with "Human" (green) and "AI Gen" (red) categories. A "Speech Transcript" section displays the text: "1 green coriander leaves, 1 chopped green ine'veave and chopped,." The "Audio Insights" section is partially visible. On the right, a "Print" sidebar shows settings for "Destination" (Microsoft Print to PDF), "Pages" (All), "Layout" (Portrait), and "Colour" (Colour). At the bottom right are "Print" and "Cancel" buttons. The top left shows the date "04/05/2026, 21:01" and the text "frontend-react". The bottom left shows "localhost:5173/#analyze" and the bottom right shows "1/2".

04/05/2026, 21:01 frontend-react

Analysis Results

Share PDF Report

Voice Authenticity
⚠ **AI_GENERATED**
Confidence Score: 59.00%

Model identified synthetic or generated patterns

Detected Language
🌐 **Tamil**
Extracted directly from Whisper multi-lingual speech transcription.

Probability Distribution

Human

AI Gen

Speech Transcript

" 1 green coriander leaves, 1 chopped green ine'veave and chopped,."

Audio Insights

localhost:5173/#analyze 1/2

Print

2 sheets of paper

Destination 🖨 Microsoft Print to PDF

Pages All

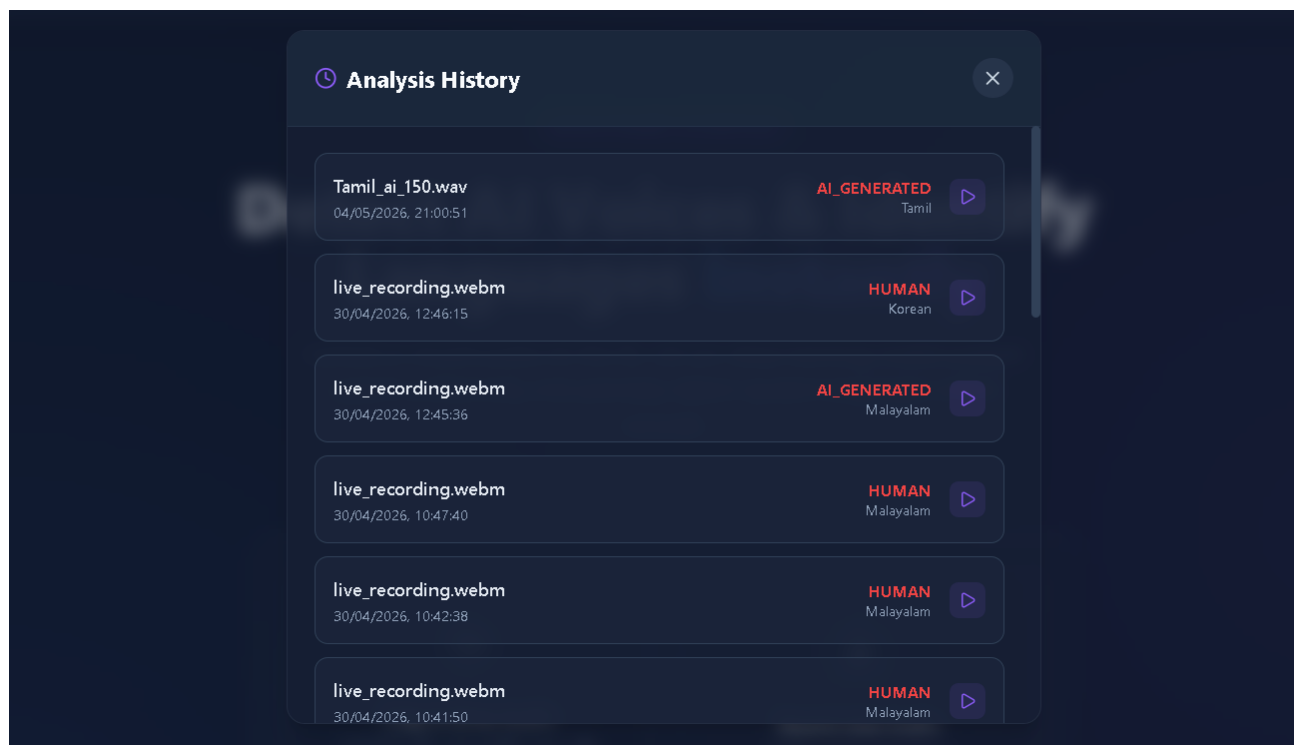
Layout Portrait

Colour Colour

More settings ⌵

Print **Cancel**

Fig 5.4 Printable PDF Report View of Audio Analysis Results



The image shows a dark-themed "Analysis History" modal window. It contains a list of six audio analysis entries. Each entry shows the filename, the date and time, the result (AI_GENERATED or HUMAN), the detected language, and a play button icon. The entries are as follows:

Filename	Date/Time	Result	Language
Tamil_ai_150.wav	04/05/2026, 21:00:51	AI_GENERATED	Tamil
live_recording.webm	30/04/2026, 12:46:15	HUMAN	Korean
live_recording.webm	30/04/2026, 12:45:36	AI_GENERATED	Malayalam
live_recording.webm	30/04/2026, 10:47:40	HUMAN	Malayalam
live_recording.webm	30/04/2026, 10:42:38	HUMAN	Malayalam
live_recording.webm	30/04/2026, 10:41:50	HUMAN	Malayalam

Fig 5.5 Analysis History Interface Showing Previous Audio Evaluations

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The proposed **AI Speech Detection and Multilingual Identification System** leverages advanced machine learning, deep learning, and audio signal processing techniques to provide an effective solution for detecting AI-generated (synthetic) speech and identifying spoken languages. By analyzing key acoustic features such as pitch variance, RMS energy variation, and zero-crossing rate, the system accurately distinguishes between human and AI-generated voices, addressing the growing challenges posed by deepfake audio and voice cloning technologies.

The integration of a **Random Forest Classifier** ensures reliable classification, while the **Whisper-based Transformer model** enhances the system by enabling accurate speech transcription and multilingual language identification. The implementation of a modern web-based application using React and FastAPI ensures real-time processing, scalability, and a user-friendly experience.

The system provides not only classification results but also confidence scores and explanatory insights, improving transparency and user trust. By combining multiple technologies into a single platform, the system enhances digital security, supports audio forensics, and helps combat misinformation caused by synthetic audio.

Overall, this project demonstrates a robust and scalable approach to speech authenticity detection and multilingual analysis, contributing to the development of safer and more reliable voice-based communication systems.

6.2 FUTURE ENHANCEMENT

Future enhancements for this system can further improve its accuracy, scalability, and real-world applicability:

- **Integration of Deep Learning Models:**

Incorporating advanced deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) can improve feature learning and enhance detection accuracy for complex audio patterns.

- **Improved Deepfake Detection:**

Developing specialized models trained on large-scale deepfake audio datasets can help the system adapt to rapidly evolving AI voice generation technologies.

- **Real-Time Streaming Analysis:**

Extending the system to support continuous real-time audio stream analysis (e.g., live calls or broadcasts) can increase its applicability in security and surveillance systems.

- **Noise Robustness Enhancement:**

Implementing advanced noise filtering and signal enhancement techniques can improve performance in noisy environments and real-world conditions.

- **Cloud Deployment and Scalability:**

Deploying the system on cloud platforms can enable large-scale processing, better resource management, and accessibility across multiple devices.

- **Multimodal Integration:**

Combining audio analysis with video or facial recognition systems can provide a more comprehensive deepfake detection solution.

- **User Feedback and Adaptive Learning:**

Incorporating user feedback mechanisms can help continuously improve model performance through adaptive learning.

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